

# Guanhua Zhang

PhD Candidate · University of Stuttgart

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## Education

### University of Stuttgart

PhD in Computer Science

• Institute for Visualisation and Interactive Systems; Supervised by Prof. Andreas Bulling

Stuttgart, Germany

Sep 2020 - Aug 2024

### International Max Planck Research School for Intelligent Systems (IMPRS-IS)

PhD Scholar

• Thesis Advisory Committee: Prof. Andreas Bulling, Prof. Steffen Staab, Prof. Katherine J. Kuchenbecker

Stuttgart, Germany

Feb 2021 - Aug 2024

### Tsinghua University

MSc in Computer Science and Technology

• Institute of HCI and Media Integration; Supervised by Prof. Yong-jin Liu

Beijing, China

Sep 2017 - Jun 2020

### Beijing University of Posts and Telecommunications (BUPT)

BSc in Computer Science and Technology

Beijing, China

Sep 2013 - Jun 2017

## Professional Experience

2020-2024 **Researcher (Wissenschaftlicher Mitarbeiterin)**, Perceptual User Interfaces group, University of Stuttgart

**Teaching Assistant**, University of Stuttgart

2015-2020 **Research Assistant**, Graphics and Geometric Computing Group, Tsinghua University

**Research Assistant**, Engineering Psychology and Human Factors Laboratory, Chinese Academy of Sciences

2014-2015 **Research Assistant**, Beijing University of Posts and Telecommunications

## Publications

### PUBLISHED

Guanhua Zhang, Zhiming Hu, Mihai Băce, Andreas Bulling. *Mouse2Vec: Learning Reusable Semantic Representations of Mouse Behaviour*, In Proc. ACM CHI conference on Human Factors in Computing Systems (CHI). 2024.

Guanhua Zhang, Matteo Bortoletto, Zhiming Hu, Lei Shi, Mihai Băce, and Andreas Bulling. *Exploring Natural Language Processing Methods for Interactive Behaviour Modelling*. In Proc. IFIP TC13 Conference on Human-Computer Interaction (INTERACT). 2023. **[Nominated for Best Doctoral Student Paper Award]**

Fang Liu, Pei Yang, Yezhi Shu, Fei Yan, Guanhua Zhang, and Yong-Jin Liu. *Emotion Dictionary Learning with Modality Attentions for Mixed Emotion Exploration*. IEEE Transactions on Affective Computing (TAFFC, IF=11.2). 2023.

Guanhua Zhang, Susanne Hindennach, Jan Leusmann, Felix Bühler, Benedict Steuerlein, Sven Mayer, Mihai Băce, and Andreas Bulling. *Predicting Next Actions and Latent Intents during Text Formatting*. In Proc. the CHI Workshop Computational Approaches for Understanding, Generating, and Adapting User Interfaces. 2022.

Guanhua Zhang, Minjing Yu, Yong-Jin Liu, Guozhen Zhao, and Dan Zhang. *SparseDGCNN: Recognizing Emotion from Multichannel EEG Signals*. IEEE Transactions on Affective Computing (TAFFC, IF=11.2). 2021. **[56 citations]**

Philipp Müller, Dominik Schiller, Dominike Thomas, Guanhua Zhang, Michael Dietz, Patrick Gebhard, Elisabeth André, and Andreas Bulling. *MultiMediate: Multi-modal Group Behaviour Analysis for Artificial Mediation*. In Proc. ACM Multimedia (MM). 2021.

Xiaobing Du, Cuixia Ma, Guanhua Zhang, Jinyao Li, Yu-Kun Lai, Guozhen Zhao, Xiaoming Deng, Yong-Jin Liu, and Hongan Wang. *An Efficient LSTM Network for Emotion Recognition from Multichannel EEG Signals*. IEEE Transactions on Affective Computing (TAFFC, IF=11.2). 2020. **[127 citations]**

Guozhen Zhao, Yulin Zhang, Guanhua Zhang, Dan Zhang, and Yong-Jin Liu. *Multi-target Positive Emotion Recognition from EEG Signals*. IEEE Transactions on Affective Computing (TAFCC, IF=11.2). 2020.

Guanhua Zhang, Minjing Yu, Guo Chen, Yiheng Han, Dan Zhang, Guozhen Zhao, and Yong-Jin Liu. *A Review of EEG Features for Emotion Recognition (in Chinese)*. SCIENTIA SINICA Informationis. 2019. [**2021 Hot Paper Award**]

Chengwen Zhang, Lei Zhang, Guanhua Zhang. *QoS-aware mobile service selection algorithm*. Mobile Information Systems. 2016.

## Awards, Scholarships, & Grants

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- 2019 **Gemalto Scholarship**, The highest amount comprehensive scholarship in department
- 2018-2019 **Vice President of Department Students' Union**  
**Outstanding Individual in Department Students' Union**
- 2014-2016 **BUPT Merit Student**, Top 3%  
**BUPT Scholarship**, Top 5%
- 2012 **2nd Prize at National Olympiad in Informatics in Provinces**, China Computer Federation

## Presentations

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### INVITED TALKS

2022 **Computational User Behaviour Modeling and Affective HCI**, Accessible & Pervasive User EXperience (APEX) Group, Hong Kong University of Science and Technology (Guangzhou) *Online*

### ORAL PRESENTATIONS

- 2023 **Exploring Natural Language Processing Methods for Interactive Behaviour Modelling**, INTERACT'23 *York, UK*
- 2022 **Predicting Next Actions and Latent Intents during Text Formatting**, CHI'22 Workshop *Online*

### POSTER PRESENTATIONS

- 2023 **Learning Reusable Semantic Representations of Mouse Behaviour**, IMPRS-IS Bootcamp *Tübingen, Germany*
- 2022 **Interact2Vec: Learning Language of Interactive Behaviour**, Stuttgart ELLIS Unit Kick-Off Event *Stuttgart, Germany*

## Teaching Experience

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### UNIVERSITY OF STUTTGART

- WS23/24 **Machine Perception and Learning**, Course co-organiser
- WS22/23 **Machine Perception and Learning**, Teaching assistant
- SS22 **Machine Perception and Learning**, Teaching assistant
- WS21/22 **Machine Learning and Computer Vision for HCI (Fachpraktikum)**, Teaching assistant
- WS20/21 **Machine Learning Approaches to Understanding User Behavior (Hauptseminar)**, Teaching assistant

## Mentoring

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### MASTER STUDENTS

- |                                                                       |                                |
|-----------------------------------------------------------------------|--------------------------------|
| <b>Yizhen Wang</b>                                                    | <i>University of Stuttgart</i> |
| Research Project: Generating Touch Typing on Mobile Devices           | <i>Jan 2024 - Jul 2024</i>     |
| <b>Yuxuan Cheng</b>                                                   | <i>University of Stuttgart</i> |
| Internship: Learning Reusable Embeddings of Mouse, Touch and Pen Data | <i>Jun 2023 - Jan 2024</i>     |
| <b>Jize Zhang</b>                                                     | <i>University of Stuttgart</i> |
| Internship: Keyboard behaviour representation learning                | <i>Apr 2023 - Jun 2023</i>     |

**Timo Hüttner, Thomas Marmann**

Course Project: Representation Learning from Mouse Based on Contrastive Learning

University of Stuttgart

Oct 2022 - Feb 2023

**Nithyasree Pushpalatha**

- Thesis: Gaze2vec: Learning Embeddings from Camera-based Eye Tracking Data
- Research Project: Hyperdimensional Computing-based Prediction of Personality Traits Using Multimodal Data
- Course Project: Predicting personality traits using multimodal data

University of Stuttgart

Nov 2020 - Dec 2022

**Xuefeng Hou**

Internship: Intent Prediction from Gaze during Text Formatting

University of Stuttgart

May 2022 - Jun 2022

**Vishnu Gangadhara Naik, Austin Vas**

Course Project: Predicting personality traits from mouse and keyboard embeddings

University of Stuttgart

Oct 2021 - Feb 2022

**Anandavalli Veerasubramanian**

Course Project: Group-Level Personality Analysis and Detection

University of Stuttgart

May 2021 - Nov 2021

**Tobias Walter, Akilandeewari Shanmuga Sundaram**

Course Project: Predicting personality traits from mouse and keyboard behaviors

University of Stuttgart

Apr 2021 - Jul 2021

**Priyadarshini Krishna Shobha**

Course Project: Predicting personality traits using multimodal data

University of Stuttgart

Nov 2020 - Feb 2021

**BACHELOR STUDENTS****Hossam Elfar**

Thesis (Co-supervise with Mayar Alfares): Private mouse and keyboard behavioral data

German University in Cairo

Mar 2023 - Sep 2023

**Manuel Schwartz**

Thesis: Generalised Encoding for Mouse and Keyboard Data

University of Stuttgart

Dec 2022 - May 2023

**Yun Ye**

Internship: Mouse behaviour representation learning

University of Stuttgart

Apr 2023 - Jun 2023

**Jiaxuan He**

Thesis: Multimodal Personality Synthesis on 3D Avatars

University of Stuttgart

Sep 2021 - Mar 2022

**Guo Chen**

Internship: EEG-based Emotion Recognition

Tsinghua University

Oct 2018 - Oct 2019

**Wanqing Chen**

Internship: EEG Signal Processing

Beijing Forestry University

Jun 2019 - Oct 2019

**Chenghan Wei**

Internship: EEG Signal Processing

BUPT

Mar 2019 - May 2019

**Academic Service****SERVICE AND OUTREACH**2023, 2021 **Co-Organiser**, Multimediate Grand Challenge at ACM MM'23 '212021 **Student Volunteer**, ACM ETRA'212018-2019 **Students' Union Vice President**, Department of Computer Science, Tsinghua University**REVIEWER**

CHI'24 '23 '22, UIST'23, MobileHCI'23, INTERACT'23, ETRA'24 '22, VRIH

**Skills****Programming:** Python, C/C++, Javascript, HTML, Node.js, SQL, Matlab, Java, Pascal; LaTeX, Markdown, Bash; Qt, Action-Script; Pytorch, Keras, TensorFlow**Languages:** Chinese (Native), English (C1), German (A2)